



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

November 27, 2025

Overview

Probability Refresher

- Definitions & Notation
- Random Numbers

Optimisation

- Common Methods
- Problems

Markov Chain Monte Carlo

- Popular Algorithms
- Tuning & Diagnosing
- Alternatives

Overview

Probability Refresher

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- Random Numbers

Optimisation

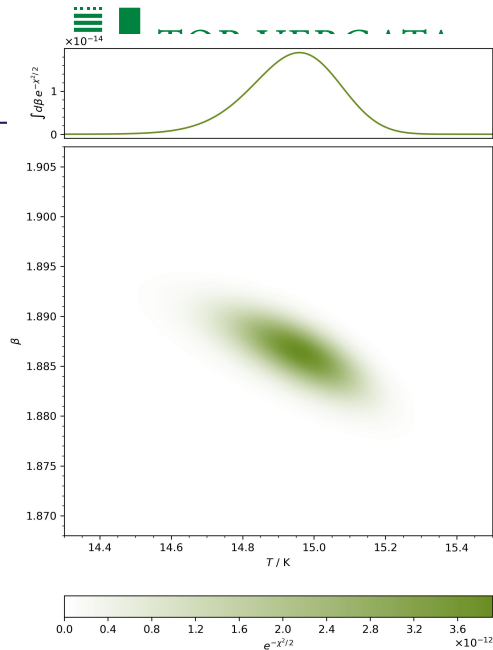
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Markov Chain Monte Carlo

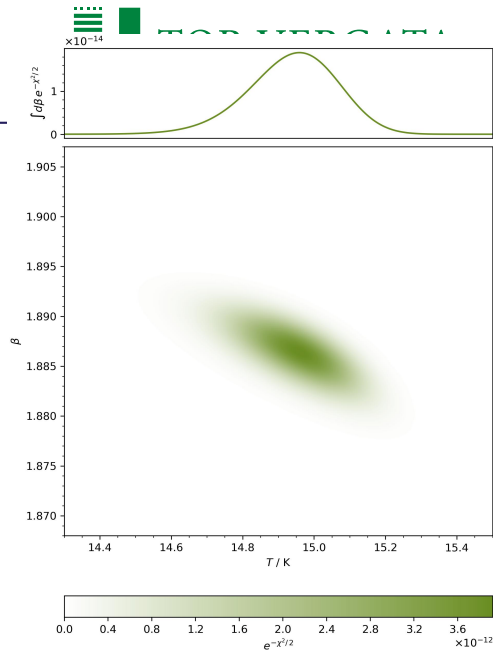
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Where I Studied

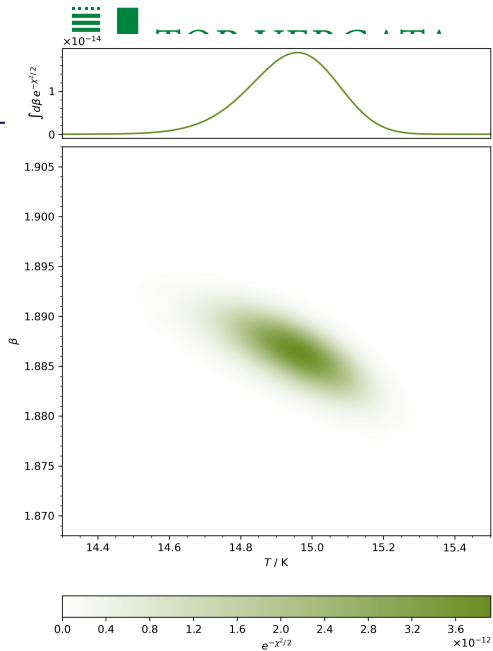
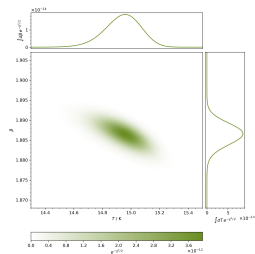
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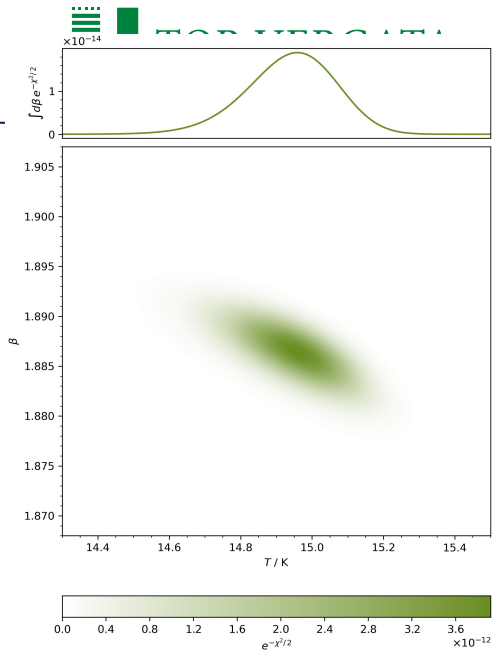
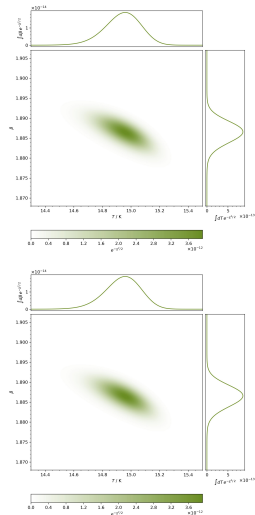
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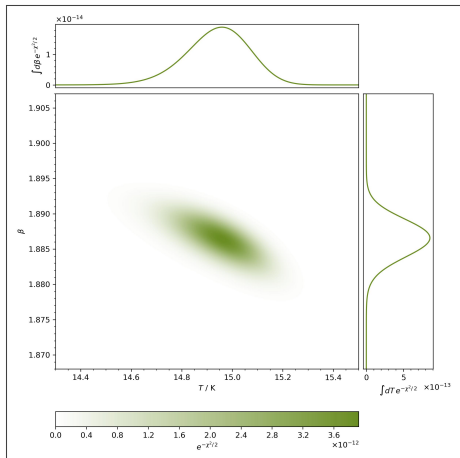


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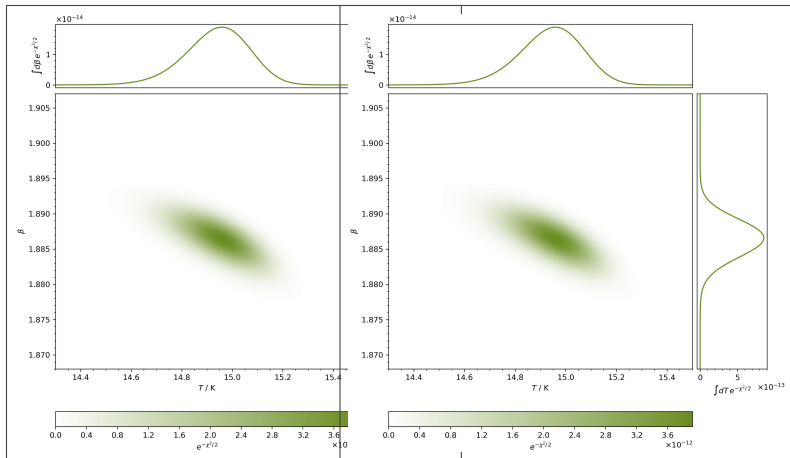


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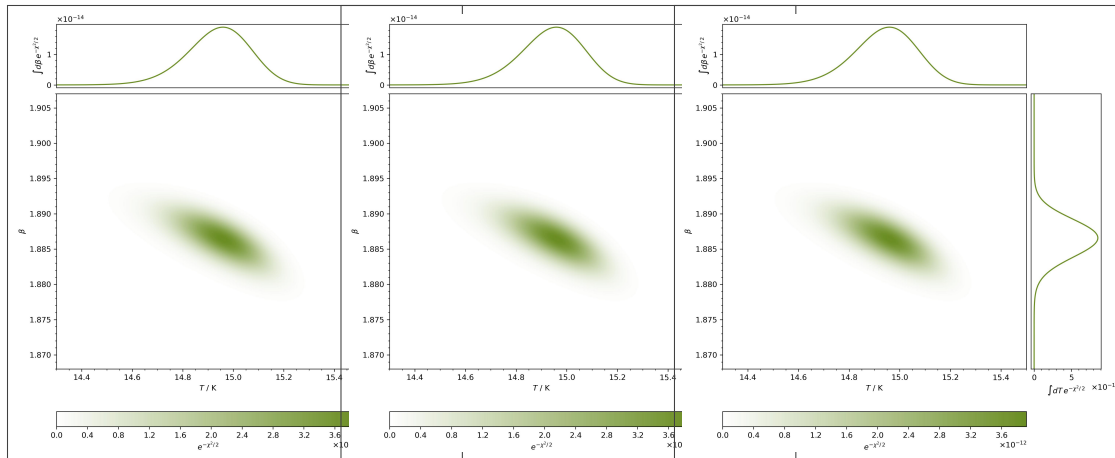
What I Studied



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Why Astrophysics



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Why Astrophysics

Interesting

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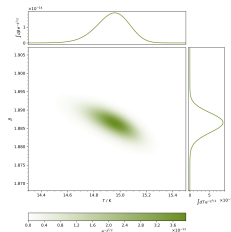
Interesting

Awesome Scales

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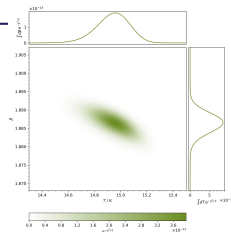
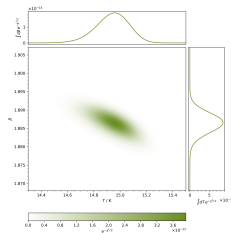
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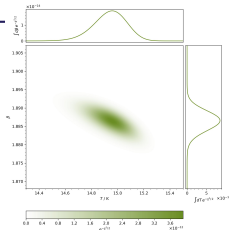
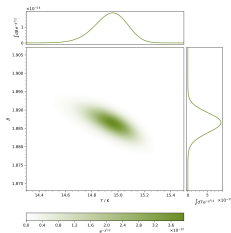


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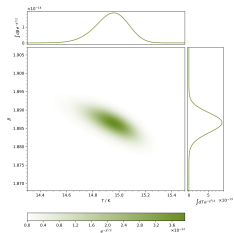
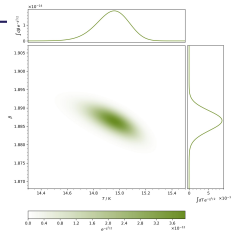
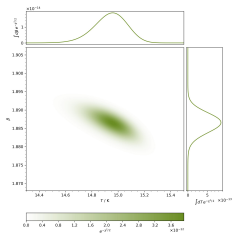


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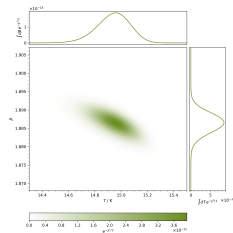
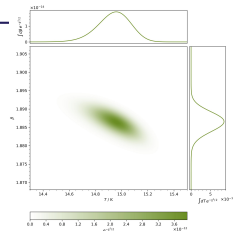
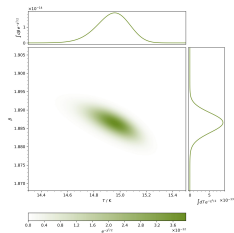
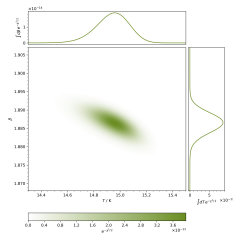


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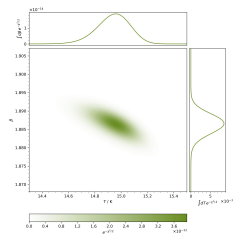
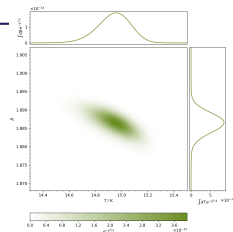
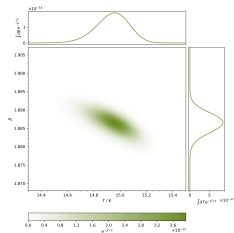
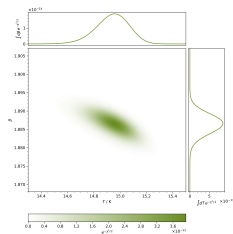
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Interdisciplinary



Why Astrophysics

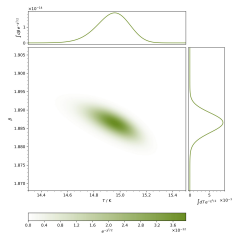
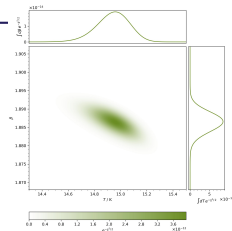
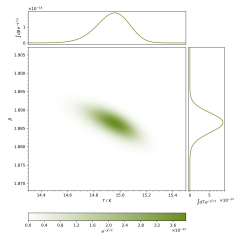
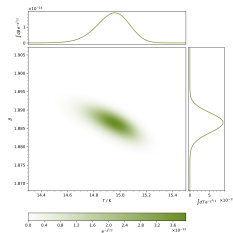
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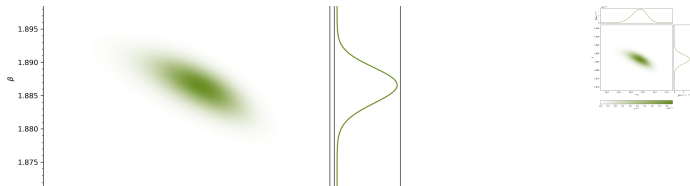
Interdisciplinary

International

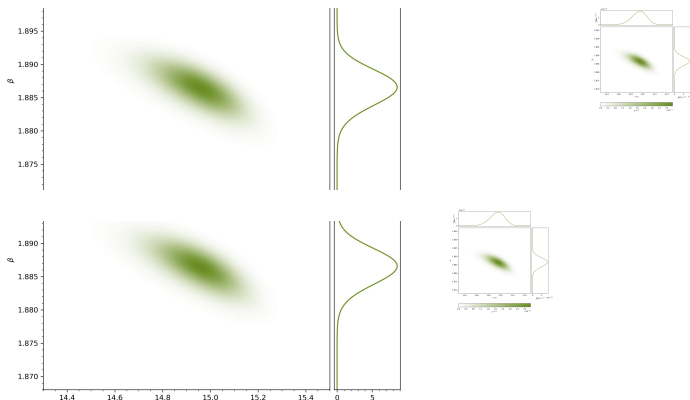


Where I Have Been

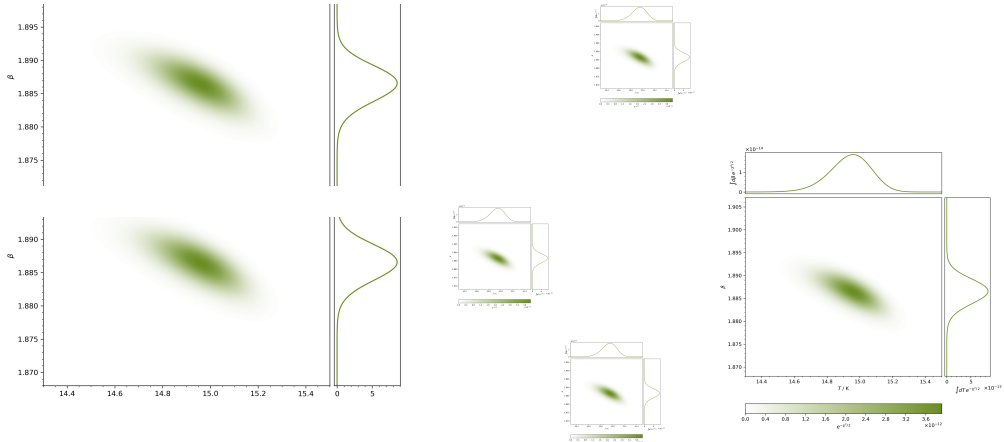
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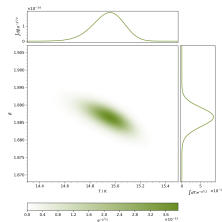
Other Interests

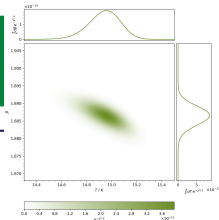
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Nature Photography

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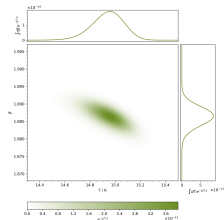
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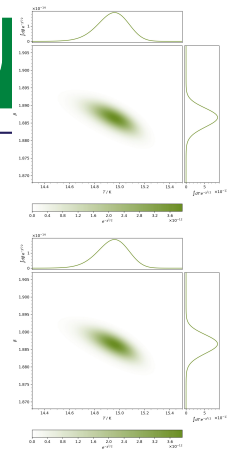
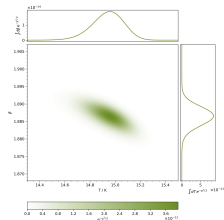
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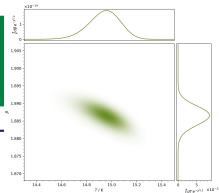
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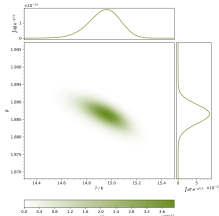
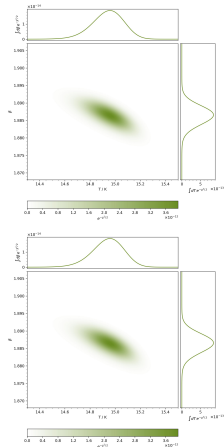
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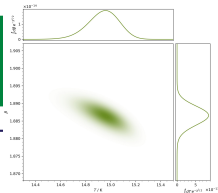




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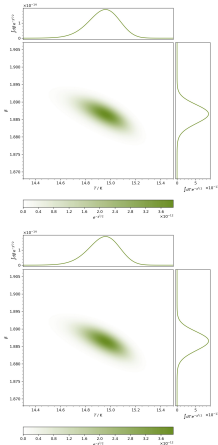
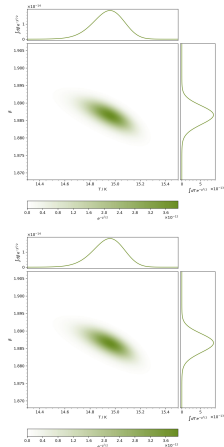
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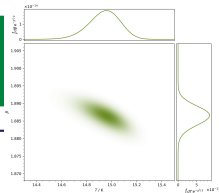




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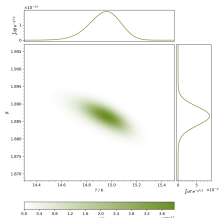
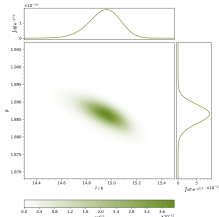
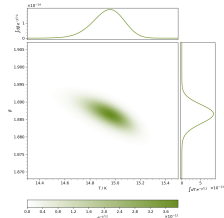
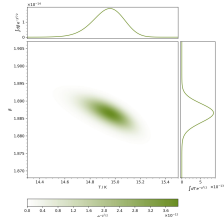


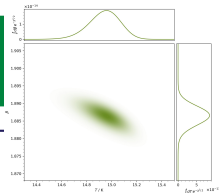


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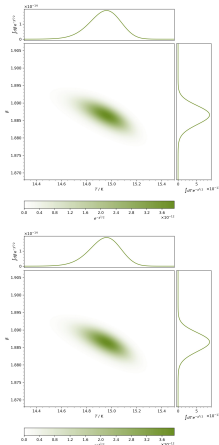
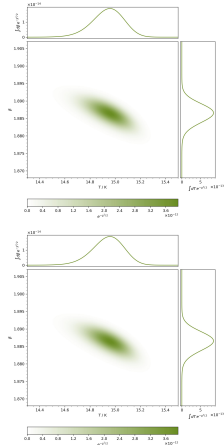


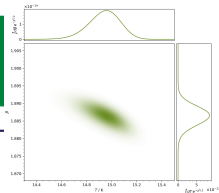
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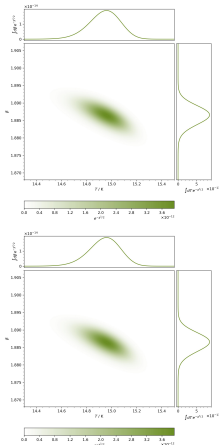
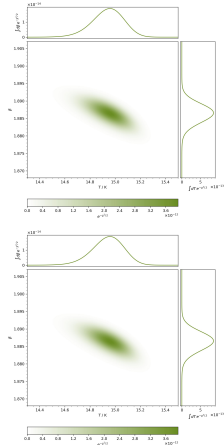
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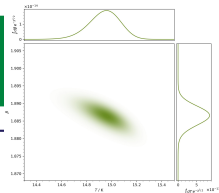
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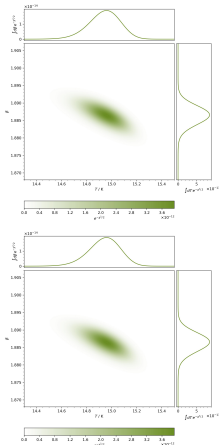
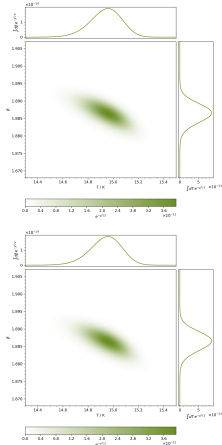
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Meeting New People



Thank You!